

CAS — Automation Assessment & Delivery SOP

From business process to governed, working automation

A concise view of the method used to determine what should be automated, where AI adds value, what should remain human, and how to implement the result reliably.

1. Discover	Map the current process end-to-end: people, systems, data, hand-offs, exceptions and pain points.
2. Identify	Find repetitive, rule-based, high-volume, error-prone or information-heavy activities.
3. Decide	Classify each activity as deterministic automation, AI-assisted, agentic, or intentionally human.
4. Architect	Define integrations, data flow, controls, security, exception handling, observability and ownership.
5. Build & Validate	Prototype the highest-value candidate, test normal and failure paths, and verify the intended outcome.
6. Govern & Evolve	Capture decisions, monitor behaviour, learn from feedback, and evolve the workflow under controlled change.

Automation Decision Framework

Pattern	Use when...	Typical control
Deterministic automation	Rules are stable, inputs are structured and the outcome is predictable.	Explicit rules, validation, retries and exception handling.
AI-assisted workflow	Unstructured content or interpretation is involved, but the workflow remains bounded.	Prompt/context controls, structured outputs, human review where needed.
Agentic workflow	The task requires multiple steps, tool calls, decisions or iterative information gathering.	Bounded tools, state/control points, observability and approval gates.
Human step	Judgement, accountability, ambiguity or risk makes automation inappropriate.	Explicit ownership and a defined hand-off rather than forced automation.

Prioritisation: Value × Feasibility × Risk

Value	Feasibility	Risk / Control
Manual effort, frequency, delay, errors, customer or employee impact.	Data quality, system access, API availability, technical complexity and change effort.	Security, privacy, reliability, explainability, exception handling and business accountability.

Decision Gates

- Automate? — Is there a repeatable activity with a clear outcome and sufficient business value?
- AI? — Does AI solve a real ambiguity or information-processing problem that deterministic automation cannot?
- Agentic? — Does the task genuinely require multi-step reasoning, tools or state, rather than a simple workflow?
- Human? — Where does judgement, approval or accountability need to remain explicit?
- Ready? — Are data, integrations, controls, monitoring and ownership sufficient for reliable operation?

CAS principle: Architecture and delivery are treated as a continuous feedback loop. Business intent, system signals, decisions, implementation and operational feedback are connected so that automation can evolve without losing architectural context or governance.

Illustrative methodology from the Continuous Architecture System (CAS), an Ideas-to-Life hands-on architecture laboratory.